



EASTMAN CHEMICAL: BOILER EFFICIENCY STUDY AND UPGRADES YIELD BIG SAVINGS

SECTOR
Industrial

LOCATION
Kingsport

FINANCIAL OVERVIEW
\$600,000

BACKGROUND

Eastman's Kingsport, TN facility dates back to the company's founding in 1920 and produces chemicals, fibers, and plastics. Taking advantage of a planned shutdown of two large boilers, the plant initiated an energy efficiency study and upgrade of two coal-fired stoker boilers in one of its powerhouses. Each boiler produces steam at 1,450 PSIG and about 900 degrees Fahrenheit, feeding topping turbo generators and the steam distribution system at the plant. In anticipation of the planned shutdown, plant personnel began to evaluate the boilers' efficiency and opportunities for performance improvements. Calculations were performed using the American Society of Mechanical Engineers' (ASME) *Performance Test Code 4 – Fired Steam Generator* standards. The results were compared to predicted performance information provided by the original equipment manufacturer (OEM). This resulted in the identification of multiple improvement opportunities that were implemented during the 2019 boiler shutdowns, resulting in an average boiler efficiency improvement of more than 4.4%. This translates into a cost savings of \$2,100,000 per year in fuel and utility costs.

The two boilers selected for improvement projects were chosen because they are the only units that run all year in that powerhouse, and they were also the least efficient topping boilers in the plant. They are permitted to burn both non-hazardous and hazardous waste. They burn non-hazardous waste from the onsite wastewater treatment plant nearly year-round. The efficiency study required approaching the problem from a variety of angles. Effectively calculating boiler efficiency showed that the units were operating well-below contract efficiency, primarily due to air leakage and high excess air levels. Time was spent with the operators investigating both their rationale for running at high excess air levels and the settings for the overfire air pressure. The OEM and onsite vessel inspectors were engaged to figure out where the air was leaking into the boiler. In addition to securing enough project funding to perform upgrades, plant personnel needed to learn how to properly rebuild the stokers. To overcome these barriers, plant personnel were able to access the company's energy efficiency improvement funding and bring in OEM technical support personnel to oversee boiler repairs.

SOLUTIONS

The projects were researched and scoped between November 2018 and March 2019, and implemented by May 2019 on the first boiler, and August 2019 on the second boiler. Project planning required design assistance and input from plant engineering, area vessel inspectors, and multiple vendors. The following improvements were made:

- Installation of upgraded stoker components and an OEM-guided overhaul resulting in better sealing of boiler 23
- Installation of new front inspection door panels
- Replacement of a slip joint on the boiler gas outlet with a fabric expansion joint
- Automation of the stoker speed control
- Overfire air pressure control
- Replacement of all access door gaskets
- Improved economizer access door sealing

The following control improvements were also executed during and directly after the boiler repairs:

- Low O₂ select control
- CO trim control
- Lower O₂ trip points
- Faster boiler draft and combustion air tuning
- New O₂ and combustion air control curves

The total cost of the boiler improvement projects was approximately \$600,000 and the efficiency of boilers 23 and 24 increased by an average of 4.43%. The upgrades have resulted in a sustained decrease in energy consumption for Boilers 23 and 24, which has yielded cost savings of \$2,100,000/year in fuel and utility costs. The annualized energy savings are based on actual coal BTU values, in-plant watt meters, ASME performance calculations, turbine steam consumption estimates, and actual in-plant steam conditions.

OTHER BENEFITS

The tuning and controls improvements allowed the boilers to run at much lower excess air levels, and air in leakage was significantly reduced by better sealing and the slip joint replacement. This lowered the power consumption requirement of auxiliary equipment, reducing steam and electricity consumption. The upgrades will result in maintenance savings as well since equipment rebuilds will be minimized. Also, the automation of stoker speed improves combustion and reduces heat in the front of the boilers, which improves the reliability of the equipment.

This project is widely applicable to any site that produces its own power or steam and has applicability for other units at the Kingsport site. The site has a total of 17 boilers and 19 turbo generators. Efficiency calculations have already been developed for the additional units and efforts are underway to scope projects for improved instrumentation. These improvements will enable better performance monitoring and the optimization of utility system loading across the plant.